

EDUCATION

University of Illinois Urbana-Champaign Ph.D. in Computer Science Research Advisor: Dr. Colleen Lewis	Urbana, IL Aug 2024–Current
University of California, Berkeley B.S. in Electrical Engineering and Computer Sciences Research Advisor: Dr. Gireeja Ranade	Berkeley, CA Aug 2020–May 2024

RESEARCH INTERESTS

Broadening Participation in Computing, Computer Science (CS) Education

PUBLICATIONS

Asterisks (*) denote co-first authorship. Daggers (†) denote mentorship.

4. **B. Agyare**, S. Jin[†], M. Acosta, C. Lewis. “Why Some Students Still Opt Out of CS: Student Perspectives in a Culturally Responsive Program.” *ACM Technical Symposium on Computer Science Education (SIGCSE)*, 2026.
3. C. Perdriau, **B. Agyare**, C. Lewis. “Extracurricular Activities Predict CS Internship Attainment.” *ACM Technical Symposium on Computer Science Education (SIGCSE)*, 2026.
2. **B. Agyare***, M. Patel*, A. Matsumoto, G. Ranade. “Broadening Participation in CS Research with Scalable Undergraduate Research Mini-Projects.” *IEEE Frontiers in Education Conference (FIE)*, 2025.
1. **B. Agyare**, A. Matsumoto, M. Patel, G. Ranade. “Student Feedback on Opt-in, Inclusive, Course-Integrated Study Groups.” *IEEE Frontiers in Education Conference (FIE)*, 2023.

WORK EXPERIENCE

IBM T.J. Watson Research Center , <i>Research Software Engineering Intern</i> • Integrated Mamba deep learning architecture into team codebase for optimized AI inference. Tools: Python, PyTorch	May 2024–Aug 2024
Microsoft , <i>Software Engineering Intern</i> • Developed storage resiliency measures for virtual storage spaces in Windows. Tools: C++, WinDbg	May 2023–Aug 2023
Google , <i>Software Engineering Intern (STEP)</i> • Built new developer features for Google’s internal code reviewing tool. Tools: Go, SQL, HTML	May 2022–Aug 2022
Lawrence Berkeley National Laboratory , <i>Research Intern</i> • Implemented graph algorithms for complex network analysis. Research Advisors: Dr. Katherine Yelick, Dr. Benjamin Brock, Dr. Giulia Guidi	May 2021–Dec 2021

AWARDS

- National GEM Consortium Ph.D. Fellowship, Sponsored by IBM Research 2024
- UC Berkeley Outstanding Graduate Student Instructor Award 2023
- Cal Alumni Association Leadership Award for Returning Students 2023

TEACHING

- **Teaching Assistant** at UIUC Siebel School of Computing and Data Science Fall 2025
 - CS 307: Modeling and Learning in Data Science
 - *Teaching and prepping weekly discussion sections for 100 undergraduate students and hosting weekly office hours.*
- **Lab Instructor** at Break Through Tech AI Program Summer 2025
 - Foundations of Machine Learning
 - *Lead nine weeks of virtual instruction and course support for 50 undergraduate students.*
 - *Hosted weekly office hours and graded lab assignments and written reports.*
- **Guest Lecturer** at UIUC Informatics Spring 2025
 - INFO 102: Little Bits to Big Ideas
 - *Gave a guest lecture on generative AI and ChatGPT.*
- **Head Teaching Assistant** at UC Berkeley EECS Spring 2022, Fall 2022, Spring 2023
 - EECS 16B: Designing Information Devices and Systems II
 - *Course covered foundational linear algebra and circuitry concepts.*
 - *Worked with professors to implement course logistics, managed a course staff of >40 TAs, and staffed office hours and online forums for course sizes of 1100, 200, and 400 students.*
- **Tutor** at UC Berkeley EECS Fall 2021
 - EECS 16B: Designing Information Devices and Systems II
 - *Staffed office hours and maintained course software infrastructure.*

POSTERS

- p3. **B. Agyare**, M. Patel, A. Matsumoto, and G. Ranade, “Broadening Participation in CS Research with Scalable Undergraduate Research Mini-Projects,” *Proceedings of the 56th ACM Technical Symposium on Computer Science Education V. 2 (SIGCSE TS 2025)*, ACM, Pittsburgh, PA, pp. 1355–1356, 2025.
- p2. M. Patel, **B. Agyare**, and G. Ranade, “Increasing Study Group Success With a Supplementary Course for Students in Gateway AI Classes,” *AAAI Symposium on Increasing Diversity in AI Education and Research*, Mar. 2024.
- p1. F. Ali, **B. Agyare**, G. Guidi, B. Brock, and K. Yelick, “Triangle Counting Algorithm with GraphBLAS,” *LBNL Computing Sciences Summer Program Poster Session*, Aug. 2021.

PRESENTATIONS AND TALKS

Panels, Presentations, and Public Speaking:

- **Panel Moderator**, Toward a more Equitable and Inclusive Future: Student Insights Oct. 2024
 - *Hosted by the Alliance for Identity-Inclusive Computing Education*
- **Student Speaker**, Joint California Summit on Generative AI May 2024
 - *Hosted by UC Berkeley CDSS, Stanford HAI, and the California Governor’s Office*
- **Panelist**, Considering Identity & Inclusion in Computing Spaces Jan. 2024
 - *Hosted by the Alliance for Identity-Inclusive Computing Education*

- **Student Speaker**, Joseph T. Gier Memorial Sculpture Dedication Ceremony
Hosted by UC Berkeley Sept. 2023
- **Presenter**, EECS Undergraduate Student Survey Presentation: Diversity in the Department
At UC Berkeley EECS Faculty Retreat Mar. 2023

Talks:

- **Understanding High School Students' Experiences Learning CS in a Culturally Responsive Computing Program**
CRA Grad Cohort for IDEALS Workshop Apr. 2025
- **Scalable Undergraduate Research Mini-Projects in a Gateway AI Course**
AAAI Symposium on Increasing Diversity in AI Education and Research Mar. 2024
- **Student Feedback on Opt-in, Inclusive, Course-Integrated Study Groups**
UC Berkeley, Algorithms & Computing for Education (ACE) Lab Dec. 2023
- **Inclusive Study Group Formation At Scale**
UC Berkeley, CS 375: Teaching Techniques for Computer Science Oct. 2022, Mar. 2023

LEADERSHIP AND OUTREACH

- **Graduate Student Ambassador**, UIUC Grainger College of Engineering 2025–Present
- **Student Advisory Board Member**, Duke's Alliance for Identity-Inclusive Computing Education 2022–Present
- **Mentor**, Graduate Society of Women Engineers (GradSWE) Undergraduate Mentoring Program 2024–2025
- **President**, Black Engineering and Science Student Association (BESSA; UC Berkeley's NSBE Chapter) 2023–2024
Previous: External Vice President (2022-2023), Pre-Collegiate Initiative Chair (2021-2022)
- **All About STEM Day Organizer**, BESSA Pre-Collegiate Programming Workshops Mar. 2021, Mar. 2022
2022: Raised \$20,000 to lead a robotics workshop for high school students in the Bay Area.
2021: Helped organize and teach a virtual data science workshop for K-12 students.

SKILLS

- **Programming Languages:** Python (NumPy, Pandas, PyTorch), Java, C++, C, Go, SQL
- **Graduate Coursework:** Natural Language Processing, Computational Social Science, Foundations of CS Education Research, Statistical Inference in Education
- **Advanced Undergraduate Coursework:** Introduction to Machine Learning; Deep Neural Networks; Efficient Algorithms and Intractable Problems; Probability and Random Processes; Optimization Models in Engineering; Principles and Techniques of Data Science; Language, Race, and Power in Education; Doing Feminist Research